

DJ Sindh Govt. Science College, Karachi

Preliminary Examination 2026

Mathematics – II

Tuesday
10-02-2026

Time allowed: 3 hours

For Science Pre-Engineering & Science General Groups

Maximum Marks: 100

Time allowed: 20 minutes

SECTION 'A' (MULTIPLE CHOICE QUESTIONS) – (M.C.Qs.)

(Marks : 20)

- NOTE (i). This section consists of 20 part questions and all are to be answered. Each question carries one mark.
(ii). The correct answer bubble must be filled on OMR sheet (I) **A B C D** pasted in answer script.
(iii). Use only blue / black ball point pen or pointer and avoid using pencil / White-o pen on OMR sheet.
(iv). All notations are used in their usual meanings. The use of Scientific Calculator is allowed.

Q 1. Choose the correct answer for each from the given options.

- (1). The Maple command $\text{diff}(5 \cdot x \cdot y, x^2)$ means
 - (a). differentiates w.r.t 'x' once
 - (b). differentiates w.r.t 'x' twice
 - (c). differentiates w.r.t 'x' thrice
 - (d). differentiates w.r.t 'x' quare
- (2). The output of the Maple command $\text{minimize}(\exp(x), x = 0 \dots 10)$ is:
 - (a). -1
 - (b). 0
 - (c). 1
 - (d). 10
- (3). The limit of the function $y = \frac{3x}{x+1}$ for $x \rightarrow \infty$ is
 - (a). 5
 - (b). -5
 - (c). $\frac{1}{5}$
 - (d). $-\frac{1}{5}$
- (4). If $f(x) = 2x$ and $g(x) = x + 1$, then $f \circ g(-5) =$
 - (a). -1
 - (b). 1
 - (c). -8
 - (d). 8
- (5). If $y = \ln(x^2 + 4)$ then $\frac{dy}{dx} =$
 - (a). $\frac{2x}{x^2+4}$
 - (b). $\frac{2}{x^2+4}$
 - (c). $\frac{x^2+4}{2x}$
 - (d). $\frac{x^2+4}{2}$
- (6). Let f be the real valued function, defined by $f(x) = \ln e^x$ then $f^{-1}(x) =$
 - (a). $\ln e^x$
 - (b). $\ln x$
 - (c). $\ln e$
 - (d). x
- (7). $x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \dots$ is the Maclaurin's series of
 - (a). $\cos x$
 - (b). $\sin x$
 - (c). $\tan x$
 - (d). $\sec x$
- (8). The point of curve which is neither point of minima nor maxima is known as point of
 - (a). failure
 - (b). inflection
 - (c). stability
 - (d). middle
- (9). The domain of $r = t^2 \hat{i} + \frac{1}{t} \hat{j} + (t+3) \hat{k}$ is
 - (a). $[0] - \mathbb{R}$
 - (b). $(0) - \mathbb{Q}$
 - (c). $\mathbb{R} - [0]$
 - (d). $\mathbb{Q} - \{0\}$
- (10). $\int x^2 \ln e^{x^2} dx =$
 - (a). $\ln e^{x^2} + c$
 - (b). $\frac{x^3}{3} + c$
 - (c). $\frac{x^3}{3} + c$
 - (d). $\frac{x^5}{5} + c$
- (11). The suitable trigonometric substitution for $\int \frac{dx}{x\sqrt{x^2-3}}$ is:
 - (a). $x = 3 \sec \theta$
 - (b). $x = \sqrt{3} \sec \theta$
 - (c). $x = 3 \sin \theta$
 - (d). $x = \sqrt{3} \sin \theta$
- (12). The equation of a straight line passes through one point (x_1, y_1) and the origin, is
 - (a). $y = \frac{y_1}{x_1} x$
 - (b). $y = \frac{x_1}{y_1} x$
 - (c). $y = \frac{y_1}{x_1} + x$
 - (d). $y = \frac{x_1}{y_1} + x$
- (13). The point $P(x_1, y_1)$ lies below the line $ax + by + c = 0$, if $x_1, y_1 \in \mathbb{R}$ and $ax_1 + by_1 + c < 0$ with
 - (a). $b = 0$
 - (b). $b < 0$
 - (c). $b > 0$
 - (d). $b = \pm 1$
- (14). If the plane intersects both nappes of the right circular cone but does not contain the vertex, the resulting intersection is a / an:
 - (a). parabola
 - (b). circle
 - (c). ellipse
 - (d). hyperbola
- (15). The equation of tangent to the circle $x^2 + y^2 = r^2$ at (x_1, y_1) is:
 - (a). $xx_1 + yy_1 = r$
 - (b). $xx_1 + yy_1 = r^2$
 - (c). $xy_1 + x_1y = r$
 - (d). $xy_1 + x_1y = r^2$
- (16). The slope of normal to the parabola $x^2 = 8y$ at $(4, 2)$ is:
 - (a). 0
 - (b). -1
 - (c). 1
 - (d). ± 1
- (17). The order and degree of the differential equation $\left(\frac{dy}{dx}\right)^3 = \sqrt{1 + \frac{dy}{dx}}$ are respectively:
 - (a). 1 and 2
 - (b). 2 and 1
 - (c). 3 and 2
 - (d). 2 and 3
- (18). The general solution of $xdy - ydx = 0$ is:
 - (a). $y = cx$
 - (b). $y^2 = cx$
 - (c). $y = cx^2$
 - (d). $y^2 = cx^2$
- (19). Let $f(x, y)$ and $g(x, y)$ are homogeneous functions of degrees 3 and 2 respectively, then the degree of $\frac{g(x, y)}{f(x, y)}$ is
 - (a). -1
 - (b). 0
 - (c). 1
 - (d). 12
- (20). In Simpson 1/3 method, the number of subinterval is the multiple:
 - (a). 4
 - (b). 1
 - (c). 2
 - (d). 3

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Mathematics – II

Tuesday
10-02-2026

Time allowed: 2 hrs. and 40 min. For Science Pre-Engineering & Science General Groups. Max. Marks: 80

NOTE: (i). Attempt any ten parts from Section 'B' and any five questions from Section 'C'.

SECTION 'B' (SHORT-ANSWER QUESTIONS) (Marks : 40)

NOTE: Answer any Ten question-parts from this section. All questions carry equal marks (i.e. 4 marks of each part).

- Q. 2
- (i). Evaluate any one of the following. (a). $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos x - \sin x}{(4x - \pi)^2}$ (b). $\lim_{x \rightarrow 0} \frac{3e^{-2x} - 5e^{-2x} + 2}{x}$
- (ii). Find the value of k so that f is continuous at $x = 2$, if $f(x) = \begin{cases} \frac{\sqrt{2x+5} - \sqrt{x+7}}{2} & , x \neq 2 \\ k & , x = 2 \end{cases}$
- (iii). If $y = \tan(a \tan^{-1} x)$ show that $(1+x^2) \frac{dy}{dx} - a(1+y^2) = 0$
- (iv). Obtain the first three terms of the Maclaurin's series for $xe^{\sin x}$.
- (v). If $\vec{x} = t\hat{i} + 2t\hat{j}$ and $\vec{y} = 2t\hat{i} + 3t\hat{k}$ are vector functions, then find $\frac{d}{dt} [\vec{x}(t) \times \vec{y}(t)]$.
- (vi). Evaluate the following definite integrals or by formula. $\int_{\frac{\pi}{4}}^{\frac{\pi}{36}} \frac{\sin \sqrt{x}}{\sqrt{x}} dx$
- (vii). Find the area, above the x -axis under the curve, between the given ordinates, $y = \ln x$, $x = 1$, $x = 3$.
- (viii). $A(2, 7)$ and $B(-4, -8)$ are coordinates of the line segment $\overline{AB}$. There are two points that trisect the segment $\overline{AB}$. Find the points of trisection.
- (ix). Find the equation of the line which passes through $(-2, -4)$ and the sum of its intercepts equal to 3.
- (x). Show that the equation of circle with center $(-g, -f)$ and touching x -axis is of the form $x^2 + y^2 + 2gx + 2fy + g^2 = 0$.
- (xi). Find the equation of the parabola, having vertex $(3, 4)$ and directrix $x = 5$.
- (xii). Find the eccentricity of the ellipse, if latus rectum is equal to half of its major axis.
- (xiii). Solve the homogeneous differential equation. $\frac{dy}{dx} = \left(\frac{y}{x}\right) + \sin\left(\frac{y}{x}\right)$
- (xiv). Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, where $f(x, y) = e^x \cos y$.

SECTION 'C' (DETAILED-ANSWER QUESTIONS) (Marks : 40)

NOTE: Answer any Five questions from this section. All questions carry equal 8 marks.

Q. 3 Evaluate any two of the following.

(a) $\int \cos(\ln x) dx$ (b) $\int \frac{\cos x}{\sin x (2 + \sin x)} dx$ (c) $\int_{-6}^{-2\sqrt{3}} \frac{dx}{x\sqrt{x^2 - 9}}$

Q. 4 (a). Differentiate with respect to " x ", $y = \ln(\sin(\ln x))$.

(b). If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots}}}$, show that $(2y - 1)y' = \cos x$.

Q. 5 A rectangular reservoir with square bottom and open top is to be lined inside with lead. Find the dimensions of the reservoir to hold $\frac{1}{2}a^3$ cubic meters, such that the lead required is minimum.

Q. 6 Find the equation of a line through the intersection of the lines $3x - 4y + 1 = 0$, $5x + y - 1 = 0$ and cutting off equal intercepts.

Q. 7 Find the center, foci, eccentricity, vertices and equation of directrices and also sketch the hyperbola. $25x^2 - 150x - 9y^2 + 72y + 306 = 0$.

Q. 8 Form the differential equation by eliminating arbitrary constants from the the given relation. $y = A \cos(\ln x) + B \sin(\ln x)$

Q. 9 Evaluate the integral $\int_0^{\frac{\pi}{2}} \sin x dx$ with 7 intervals by using Trapezoidal rule.

Good Luck

Handwritten notes:

$y = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) + y_1$

$y = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) + y_1$

(x, y)

Handwritten notes:

$X = \frac{m_1 - m_2}{1 + m_1 m_2}$

$Y = \frac{m_2 - m_1}{1 + m_1 m_2}$